

Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

POSTLAB EXERCISES (13 points)*Submit the postlab to the TA at the end of the lab.***Measurement 1: Force vs. Separation (2 points)**

Coulomb's Law applies to point charges, but we use spheres of radius a . When the spheres are far apart ($r \gg a$), they behave like point charges because the charge on each is distributed nearly uniformly. When the spheres are close together ($r \approx a$), the charges bunch toward each other, leading to non-point-like behavior. In a more advanced course you will learn how to calculate this effect, but for now it suffices to know that the corrected observable is

$$\Delta\theta_c = \frac{\Delta\theta}{\beta}, \quad \text{where} \quad \beta = 1 - 4 \left(\frac{a}{r} \right)^3 \quad (6)$$

is the correction factor. Record your measurements of $\Delta\theta$ for each separation r in Table 1 and compute the mean of your N measurements:

$$\overline{\Delta\theta} = \frac{1}{N} \sum_{i=1}^N \Delta\theta_i. \quad (7)$$

Then compute the correction factor β and the corrected mean twist angle $\overline{\Delta\theta}_c$ using eq. (6). Finally, for each separation r , tabulate $1/r^2$, $\ln(r/a)$, and $\ln \overline{\Delta\theta}_c$. You will plot the final two columns on the next page.

Table 1: **Measurement 1 Data**

r [cm]	$\Delta\theta$	$\overline{\Delta\theta}$ [°]	β	$\overline{\Delta\theta}_c$ [°]	$1/r^2$	$\ln(r/a)$	$\ln(\overline{\Delta\theta}_c)$
20							
14							
10							
9							
7							
5							

Question 4**2 points** Measurement**2: Force vs. Charge (11 points)**

Table 2: Measurement 2 Data

ΔV_2 [kV]	$\Delta\theta$	$\overline{\Delta\theta}$ [°]	β	$\overline{\Delta\theta}_c$ [°]	$\ln(\overline{\Delta\theta}_c)$	$\ln(\Delta V_2/1 \text{ kV})$
6						
5						
4						
3						
2						
1						

Question 5**2 points****Question 6****2 points**

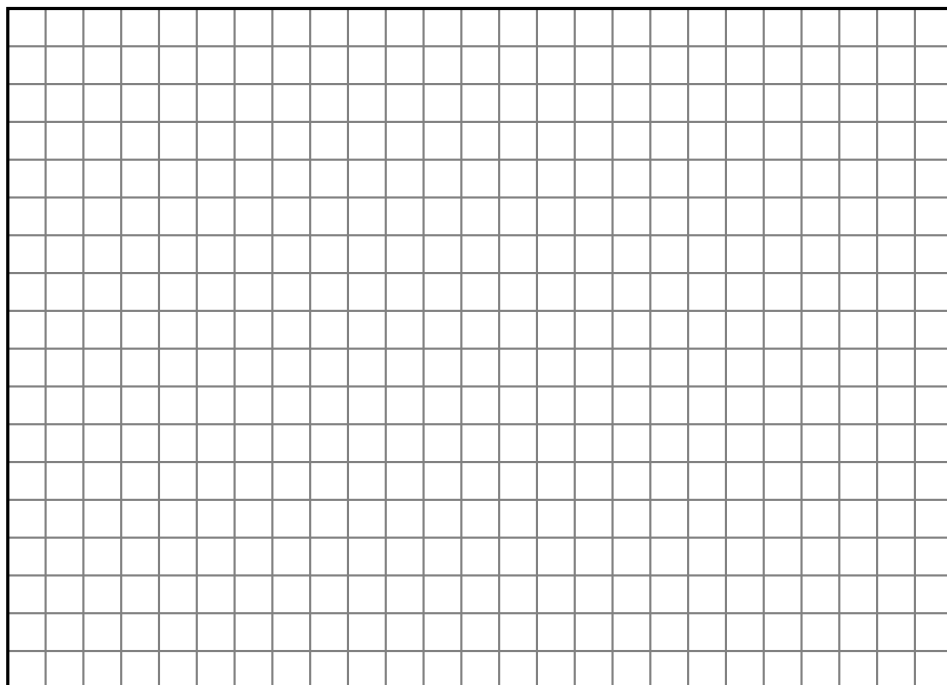
Show one complete example of how you corrected your data. Choose any one row from Table 1 and compute β and $\overline{\Delta\theta}_c$ step by step.

Question 7**2 points**

Using your log-log plot from Question 6, calculate the slope of the best-fit line. Show your calculation by identifying two points on the line.

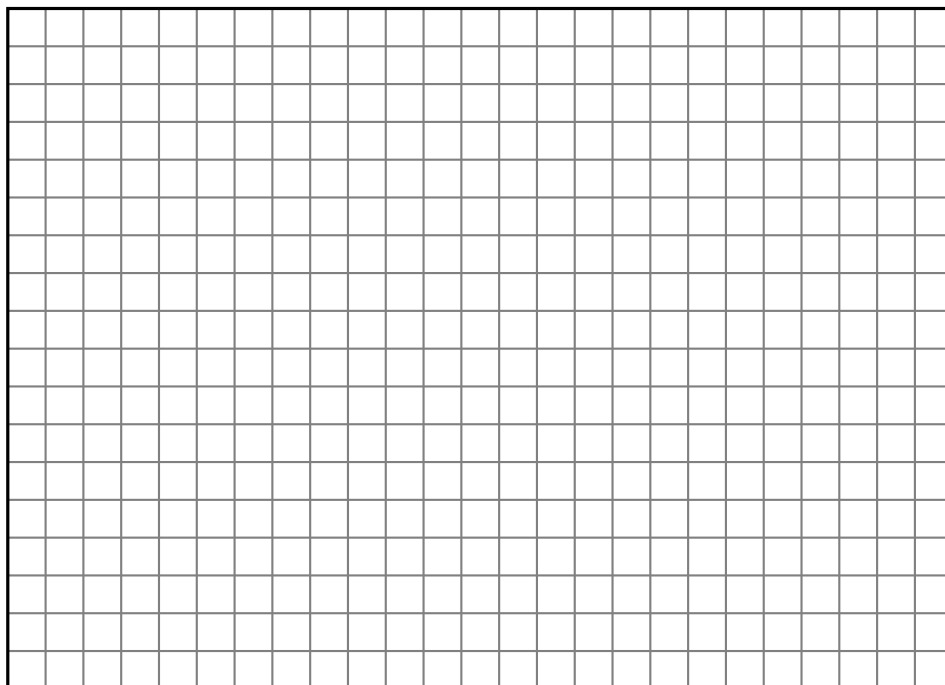
Question 8**1 point**

Make a scatter plot of $\ln(\overline{\Delta\theta_c})$ versus $\ln(r/a)$ with the independent variable on the horizontal axis and the dependent variable on the vertical axis. Label your axes.



Question 9**1 point**

Make a scatter plot of $\ln(\overline{\Delta\theta_c})$ versus $\ln(\Delta V_2/1 \text{ kV})$ with the independent variable on the horizontal axis and the dependent variable on the vertical axis. Label your axes.

**Question 10****1 point**

For both graphs, add a dashed line that fits the data well. Use each plot to estimate the slope m and y -intercept b , and write down the equation of the best-fit line in the form $y = mx + b$.

Question 11**2 points**

Analyze the best-fit lines from both plots. What mathematical relationship exists between $\ln(\overline{\Delta\theta_c})$ and $\ln(r/a)$? What relationship exists between $\ln(\overline{\Delta\theta_c})$ and $\ln(\Delta V_2/1\text{ kV})$? Do these relationships match what Coulomb's Law predicts? If your measured slopes differ from the expected values, identify possible sources of experimental error.